

Oluwatobi Melvyn Mayungbo

Data Scientist, Statistics and Forecasting

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SUMMARY

Data scientist with a B.Sc. in Statistics who builds probabilistic forecasts and tests them honestly. Built a football predictor backtested gameweek by gameweek over 1,050 matches with calibrated probabilities, quantile demand forecasts validated with rolling-origin backtesting, and fraud models whose calibration error fell from 0.0063 to 0.0005.

SKILLS

Statistics and Causal Inference: Time-Series Backtesting, Statistical Inference, Logistic Regression, Experiment Design, Power Analysis, CUPED, Causal Inference (S/T/X-Learners, Causal Forests), Uplift Modeling, Qini and AUUC, WOE Scorecards

Machine Learning: LightGBM, Model Calibration (Isotonic Regression, Temperature Scaling), Feature Engineering, Scikit-learn, PyTorch, XGBoost, Deep Learning, Sequence Models (GRU, TCN, Transformers), SHAP

Engineering and MLOps: Python, SQL, DuckDB, PostgreSQL, Redis, FastAPI, Docker, GCP Cloud Run, AWS ECS Fargate, Railway, Prefect, dbt-core, MLflow, Evidently, CI/CD (GitHub Actions), Git, Pytest, Streamlit, React

Data Analysis: Pandas, NumPy, SQL, Anomaly Detection

AI and LLM Systems: Retrieval-Augmented Generation (RAG), Hybrid Search (BM25, Dense Embeddings, RRF), LLM Evaluation and Guardrails, Prompt Engineering, Citation Verification, Model Context Protocol (MCP), Hugging Face, ONNX Runtime

PROJECTS

Premier League Match Predictor | [GitHub](#)

Tech: Python, LightGBM, scikit-learn, pandas, SQLite, Flask, Vercel

- Built a predictor that forecasts Premier League outcomes and scorelines each gameweek and retrains itself after every gameweek, keeping a public record of every past prediction.
- Blended a LightGBM classifier over a 204-column feature table with a multinomial logistic regression, trained on log loss so draws keep realistic probability.
- Backtested gameweek by gameweek over 1,050 matches: 52.1% outcome accuracy against 43.2% for always picking the home side, and log loss 0.9948 against 1.0061 for an Elo-only baseline, with well calibrated probabilities.

Self-Operating Demand Forecasting Platform | [GitHub](#)

Tech: Python, Prefect, dbt-core, DuckDB, LightGBM, MLflow, Evidently, FastAPI, Docker, GitHub Actions

- Built an end-to-end hourly demand forecasting system for NYC taxi zones, producing 24-hour quantile forecasts through an orchestrated Prefect pipeline spanning ingestion, dbt/DuckDB transformations, backtesting, drift monitoring and conditional retraining.
- Modelled asymmetric business cost by penalising unmet demand at 3× idle supply and shipping the cost-optimal quantile rather than the median, cutting expected cost by 25% on an identical model and beating a seasonal-naive baseline by 32%.
- Validated with rolling-origin backtesting across 6 expanding-window folds against seasonal-naive and historical-mean baselines, reporting MASE and pinball loss, with automated leakage tests that fail the build if any feature reads past the forecast origin.

Sequence-Based Transaction Fraud Detection System | [GitHub](#)

Tech: Python, PyTorch, LightGBM, ONNX Runtime, FastAPI, Docker, SHAP

- Handled extreme class imbalance with focal loss and weighted sampling and isolated their failure modes: focal loss degraded probability calibration (ECE 0.0063) while weighted sampling cost 0.042 PR-AUC. Corrected calibration with temperature scaling and isotonic regression, reducing ECE to 0.0005.

EXPERIENCE

Machine Learning Instructor, SQI College of ICT

March 2026 to Present

- Teach machine learning and deep learning to student cohorts, covering statistical foundations, classical ML algorithms, neural architectures, sequence models and production evaluation metrics.
 - Instruct students on data analysis, feature engineering and model building with PyTorch, Scikit-Learn, Pandas and NumPy.
- Internal Auditor, NISER** May 2024 to July 2024
- Analysed financial and operational records to identify inconsistencies and irregularities, performing manual anomaly detection across complex transactional datasets.

EDUCATION

- B.Sc. Statistics, University of Ibadan** January 2021 to March 2025
- Professional Diploma in Artificial Intelligence, SQI College of ICT** July 2025 to Present

CERTIFICATIONS

- Google Data Analytics Professional Certificate
- Google Advanced Data Analytics Professional Certificate
- Claude Code In Action